

Mini-Assignment 1:

Stress Testing Models with Ambiguous Sentences

Course: Computational Linguistics – I

Deadline: January 9, 2026 | 23:59

Objective

The objective of this assignment is to understand how modern language processing systems handle **ambiguity and context**, and to reflect on why such phenomena remain challenging for machines. This assignment emphasizes **linguistic analysis and reasoning**, not programming.

Task Description

You will analyze ambiguous sentences using two widely used systems:

- a machine translation system (Google Translate)
- a large language model (ChatGPT or Gemini)

Your goal is not to rank these systems, but to understand **what kinds of knowledge they appear to use or miss** when dealing with ambiguity.

Part A: Creating Ambiguous Sentences

Select **any Indian language** you are comfortable with (Hindi, Gujarati, Telugu, Tamil, Kannada, etc.). You must write examples and provide **glossed representations**.

Create:

- **Two lexically ambiguous sentences**
- **Two structurally (syntactically) ambiguous sentences**
- **Two semantically ambiguous sentences**
- **Two pragmatically ambiguous sentences**

For each sentence:

- write the sentence clearly
- provide a gloss (as shown below)
- briefly describe the two possible interpretations

(Total: 8 sentences)

Glossing: Format and Example

When giving Indian language examples, use the following four-level format:

Level	Representation
Original Sentence	राम ने दूरबीन से आदमी को देखा।
Transliteration	raam ne doorbeen se aadmi ko dekhaa
Gloss	Ram ERG telescope with man ACC saw
English	Ram saw the man with a telescope.

Two Possible Interpretations

- **Instrument Interpretation:** Ram used a telescope to see the man.
- **Modifier Interpretation:** Ram saw a man who was carrying a telescope.

This example shows how ambiguity arises from **structural attachment**, not from vocabulary alone.

Part B: Tool-Based Analysis

For each sentence:

1. Run the sentence through **Google Translate** and observe whether the translation:
 - preserves one interpretation, or
 - ignores the ambiguity altogether
2. Ask an **LLM (ChatGPT or Gemini)** the following:

“Explain all possible interpretations of this sentence.”

Part C: Comparison Table

Prepare a comparison table like the one below (add rows as needed):

Sentence	Type of Ambiguity	Google Output	Translate	LLM Explanation	Your Observation
Sentence 1	Lexical / Structural / Pragmatic				
Sentence 2	Lexical / Structural / Pragmatic				

In the **last column**, clearly write your own linguistic analysis.

Part D: Reflection (150–200 words)

Write a short reflection addressing the following:

Discuss one sentence from your analysis where ambiguity resolution was non-trivial. If the systems struggled to resolve cultural or pragmatic meaning, explain why this is difficult for machines. If the systems succeeded, explain what kind of knowledge (linguistic, world, or contextual) they appear to be using, and what limitations may still remain.

Submission Instructions

- Submit **one single PDF file**
- File name format:

`CL1_MiniAssignment1_<RollNumber>.pdf`
- Clearly label sections as Part A, Part B, Part C, and Part D
- Handwritten submissions are allowed only if scanned clearly

Important Notes

- This is a **conceptual assignment**; no programming is required.
- You may use examples from books, newspapers, or your own creation. Please give proper citation if applicable.
- Do not use LLMs to generate answers without analysis.
- Clarity of reasoning matters more than length or sophistication.